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Patellofemoral Pain

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Patellofemoral pain is common, with an incidence as high as 50% in some populations.[1] Often referred to as anterior knee pain, patellofemoral pain can be challenging for both patients and clinicians, as it can often be difficult to determine the underlying etiology responsible for the pain. Some patients with patellofemoral pain have no definable pathology. This process has been described by different names, including patellofemoral pain syndrome (PFPS) and idiopathic anterior knee pain. It is important to note that these terms to some extent are diagnoses of exclusion and should not be used to mislabel a definable disease process. It is also a misnomer to attribute all patellofemoral pain to "chondromalacia," because in many cases, no true chondral pathology exists.

ANATOMY

The anterior aspect of the knee has several unique properties that serve as a predisposition for pain. The patellofemoral joint itself is a sliding articulation with six degrees of freedom. Patellar tracking is defined by translation (medial/lateral), glide (superior/inferior), and tilt (medial/lateral). During initial knee flexion, the patella translates approximately 4 mm medially to engage the trochlear groove at approximately 20 degrees of knee flexion. With progressive flexion, the patella translates approximately 7 mm laterally by the time the knee reaches 90 degrees of flexion, associated with approximately 7 degrees of lateral tilt. Maximum contact between the trochlear groove and the patella occurs at 45 degrees of knee flexion.[126]

In addition to the bony anatomy described above, there are both passive and dynamic restraints that contribute to the stability of the patellofemoral joint. Passive restraints include the osseous morphology, the medial patellofemoral ligament (MPFL), the medial patellomeniscal ligament, and the lateral retinaculum. Dynamic restraints include the vastus medialis oblique (VMO) and the vastus lateralis. A thorough understanding of these structures allows for a better ability to understand the common clinical problems associated with the patellofemoral joint as it attempts to meet its contradictory demands of providing both mobility and stability.[122–127] Any imbalance to this complex network of structures can result in pain, instability, or both.

The patella serves to improve the mechanical leverage of the quadriceps muscle by increasing the lever arm and transmitting the tensile forces generated by the quadriceps to the patellar tendon. There is a delicate interplay of forces acting on the patella, which sees an incredible amount of force.[121] In particular, the patellofemoral joint experiences joint reaction forces that become higher as the knee flexion angle increases (Fig. 106.1).[2] Squatting and jumping, for example, have been reported to create forces 7.6-times and 12-times body weight, respectively.[3,4]

The anterior aspect of the knee also provides significant sensory feedback. In a 1998 study by Dye et al., in which SF Dye (first author of that study) underwent bilateral knee arthroscopies without anesthesia in an effort to determine conscious neurosensory characteristics of the knee, the authors described the highest subjective pain sensations in the anterior synovium, fat pad, and joint capsule.[5] Interestingly, structures that commonly require intervention (e.g., the menisci, articular cartilage, and ligaments) were much less sensitive.[5] Dye and colleagues[5] also reported accurate spatial localization to palpation in the anterior structures (Fig. 106.2), which correlates with previous findings about the relatively high concentration of type IVa afferent nerve fibers in the patellar ligament and retinaculum.[6]

Malalignment is a commonly proposed mechanism for patellofemoral pain.[10,11] Overall limb malalignment can cause or exacerbate patellofemoral joint disorders. The lines of action of the quadriceps and the patellar tendon are not collinear—the angular difference between them is called the quadriceps (Q) angle (Fig. 106.3). The Q-angle is an angle formed from a line drawn from the anterior superior iliac spine to the middle of patella, and from the middle of the patella to the tibial tuberosity on a full-length weight-bearing radiograph. A normal Q-angle in males is 13 to 14 degrees and 17 to 18 degrees in females.[120] Because of this angle, the force generated by the quadriceps both extends the knee and pulls the patella laterally. The relative magnitude of this laterally directed force is proportional to the Q-angle. External rotation of the tibia, internal rotation of the femur, and increasing knee valgus each result in an increase in Q-angle and an associated increase in the laterally directed force within the patellofemoral joint. While patella maltracking as a result of any of the above-listed factors can cause substantial pain and dysfunction, some authors have questioned the link between malalignment and patellofemoral pain.[12–14] The common occurrence of pain at rest in patients with patellofemoral pain is an argument against malalignment as a major component of anterior knee pain.[15] Dye[12] proposed that pain is from a loss of tissue homeostasis. Increased intraosseous pressure in the patella with resultant tissue ischemia has also been proposed.[16] Other studies suggested altered vascular flow (arterial and venous) and

Abstract

Anterior knee pain is one of the most common musculoskeletal complaints seen in emergency rooms, primary care offices, and orthopaedic clinics. A thorough understanding of the anatomy, biomechanics, and potential pathology of the patellofemoral joint is necessary given this joint's unique characteristics. Evaluation includes a detailed history, physical exam, provocative tests, and imaging. Patellofemoral pain syndrome and idiopathic anterior knee pain are diagnoses that are used interchangeably and are diagnoses of exclusion. Patellofemoral pain encompasses multiple potential diagnoses which can be difficult for the clinician to sort out; an organized system for evaluation and treatment is necessary for efficient and thoughtful patient care.

Keywords

patellofemoral
knee
Q angle
idiopathic anterior knee pain
patella
patellofemoral pain syndrome

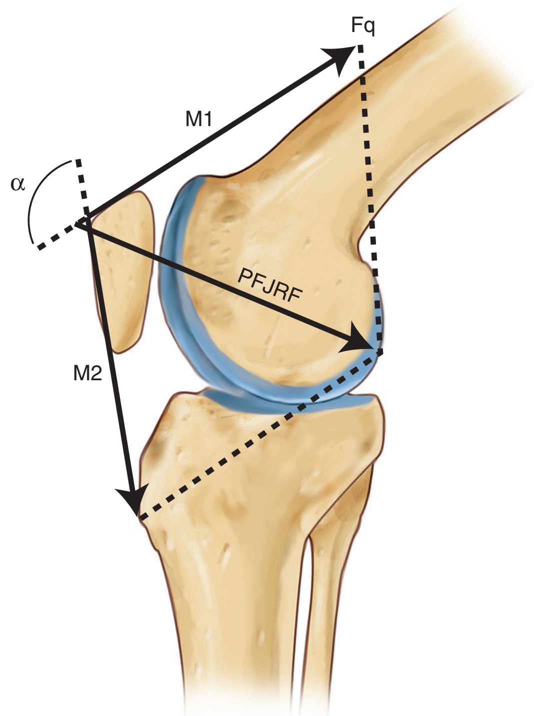

Fig. 106.1 The patellofemoral joint reaction force *(PFJRF)* becomes higher as the knee flexion angle increases. In complete extension, *M1* and *M2* are in opposite directions, but in the same plane; the resultant PFJRF is almost zero. As flexion increases, M1 and M2 converge, and the vector PFJRF increases. (From Scott WN, ed. *Insall & Scott Surgery of the Knee.* 5th ed. Philadelphia: Elsevier; 2012.)

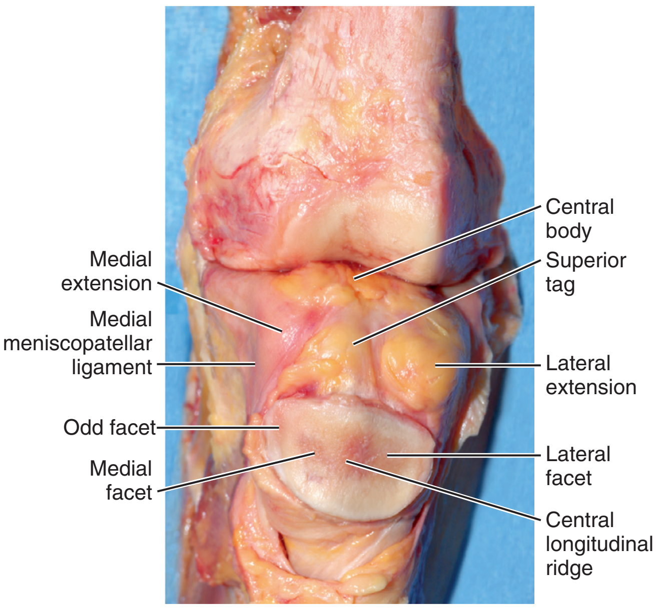

Fig. 106.2 Posterior anatomy of the patella with adjacent capsular thickenings and fat pads with the medial and lateral meniscopatellar ligaments. (From Noyes FR, ed. *Noyes' Knee Disorders: Surgery, Rehabilitation, Clinical Outcomes.* Philadelphia: Elsevier; 2010.)

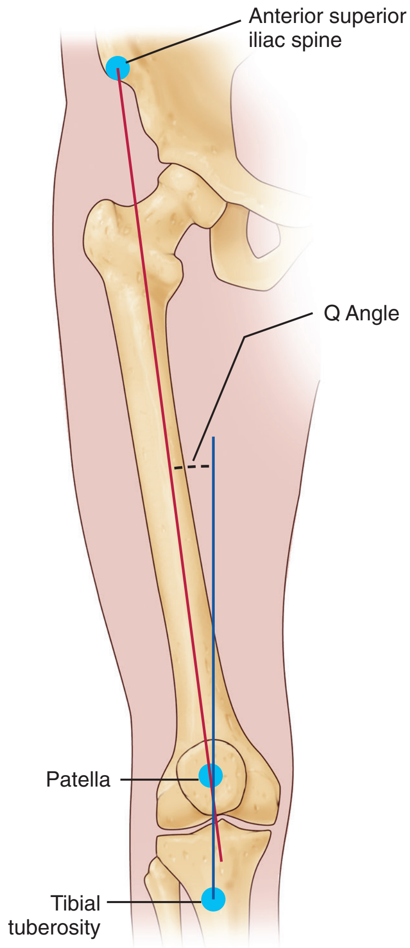

Fig. 106.3 Q angle. (Modified from Livingston LA, Mandigo JL. Bilateral Q angle asymmetry and anterior knee pain syndrome. *Clin Biomech [Bristol, Avon].* 1999;14[1]:7–13.)

concomitant degenerative conditions as being associated with symptoms generation in patellofemoral syndrome.[17–19]

PATIENT EVALUATION

History

A comprehensive history is critical to evaluation of patients presenting with patellofemoral pain. The first distinction is determining if the patient is presenting with pain or if the patient is complaining of patellar instability. While in many cases, patients have both of these complaints, when asked about their primary complaint, patients will often endorse either a chief complaint of instability with reasonable pain resolution after an acute incident, or significant pain with no symptoms of true instability. Many of the diagnoses associated with acute traumatic injuries (e.g., extensor mechanism disruption, patellar fractures, and patellar instability) are addressed in other chapters. Chronic, activity-related anterior knee pain is a common clinical complaint that occurs in a variety of knee disorders and is associated with a broad differential diagnosis (Table 106.1).

Patients should be asked about activities that cause pain, if pain occurs at rest and/or during sleep, and if the pain is

TABLE 106.1 Differential Diagnosis of Patellofemoral Pain

Acute Pain	Chronic Pain	Constant Pain

• Extensor mechanism disruption
• Patellar instability (dislocation, subluxation)
• Patella fracture
• Chondral injury
• Osteochondral injury
• Loose body
• Infection
• Hip pathology (referred pain)
• Slipped capital femoral epiphysis (referred pain)
• Lumbar spine referred pain
	• Idiopathic (patellofemoral pain syndrome)
	  • Malalignment
	  • Muscular imbalance/atrophy
	  • Overuse injury
	• Patella chondromalacia
	• Synovial fat pad impingement
	• Symptomatic plica
	• Intraosseous hypertension of the patella
	• Lateral patella compression syndrome
	• Osteoarthritis
	• Inflammatory arthritis
	• Iliotibial band syndrome
	• Patellar tendonitis
	• Osteochondritis dissecans
	• Osteochondroses
	  • Osgood-Schlatter
	  • Sinding-Larsen-Johansson
		• Complex regional pain syndrome
		• Neuroma
		• Oncologic process

associated with swelling and/or mechanical symptoms. A history of effusions often is associated with an intra-articular problem, such as a chondral lesion, whereas the absence of effusions may suggest an extra-articular etiology of pain. Details on prior successful, as well as unsuccessful treatments, including activity modification, physical therapy, injections, and/or surgeries, should be documented. In the setting of chronic, activity-related anterior knee pain without a history of trauma, patients will often complain of discomfort while ascending or descending stairs, during deep squats, or after sitting for prolonged periods. In addition, it can be common for patients to complain of symptoms bilaterally.[20] An onset of patellofemoral pain is also common during sudden periods of increased activity, such as during high-intensity training with military recruits.[21]

Physical Examination

Similar to a comprehensive history, a thorough physical examination is essential in the evaluation of patients with patellofemoral pain. Notably, it is important to appreciate that many variations exist with respect to patellofemoral anatomy. These variations can become evident during examination, and while for some patients, they may be responsible for patellofemoral pain, in other cases, they may simply be "normal" for that particular patient. The examination findings are a portion of the complex clinical picture commonly encountered in the patient with anterior knee pain.

General

Patients presenting with anterior knee pain should undergo a standard physical examination of both knees, including an assessment of gait and alignment, range of motion (ROM), strength, generalized ligamentous laxity, stability, and neurovascular status. In addition, the clinician should examine the ipsilateral hip, particularly in pediatric and adolescent patients, as primary hip pathology can often present as knee pain.

Inspection

While there are many different techniques for performing a physical examination of the knee, it can be efficient to begin with the patient standing, then seated, and then supine (and prone, if desired). The critical aspect is ensuring that all components of the examination are completed for every patient. If beginning with the patient standing, the clinician can assess limb alignment, rotation profile, and gait. Patients with increased femoral anteversion, genu valgum, and pes planovalgus may be at an increased risk for patellofemoral pain. In addition, patients with external tibial torsion and foot pronation may experience increased patellofemoral pain. Monitoring the patient's gait as well as observing a single-leg squat can be helpful in the assessment of patellar tracking and general core strength.

Once seated with the legs hanging off the end of the examination table, an evaluation of patellar tracking and an assessment of the Q-angle can be performed. Techniques for assessing Q-angle are variable, and can be performed with the patient in the standing, seated, or supine position. In addition, Q-angle measurements can be performed statically (static Q-angle) with the quadriceps contracted or with the quadriceps relaxed, as well as in a dynamic-fashion (dynamic Q-angle). It is important to appreciate that the Q-angle value may be affected by the relationship of the proximal femur and the amount of knee flexion (Figs. 106.4 and 106.5), as well as by the method in which the assessment is performed. The Q-angle assessment with the knee in flexion may be more accurate than with the knee extended, as in flexion the patella is well-seated in the trochlear groove, whereas in extension, a laterally translated patella may falsely result in a "normal" Q-angle. Some authors have suggested a dynamic Q-angle evaluation may be of more utility than a static measurement in the evaluation of patients with patellofemoral pain, though this is controversial.[21a] An elevated Q-angle may predispose the patient to anterior knee pain, although significant variability exists. In the seated position, asking the patient to

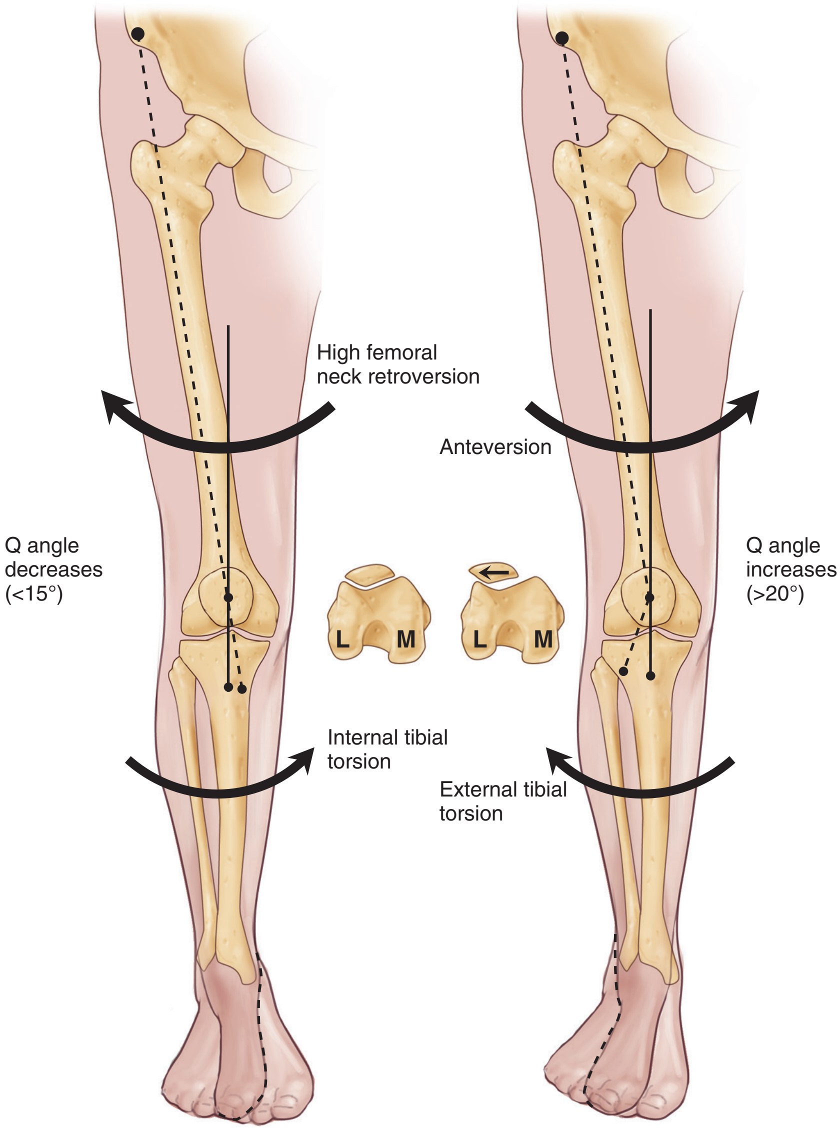

Fig. 106.4 High femoral neck retroversion rotates the distal end of the femur externally. In combination with internal tibial torsion, the Q angle is decreased. Patellar tracking is improved, and patellofemoral sulcus alignment is normal. High femoral neck anteversion rotates the distal end of the femur internally. In combination with external tibial torsion, the Q angle is increased. Patellar tracking is compromised, and the patella tends to track laterally. (From Tria Jr AJ, Klein KS. *An Illustrated Guide to the Knee*. New York: Churchill Livingstone; 1992.)

actively flex and extend his/her knee will allow for an assessment of patellar tracking. A J-sign is noted when the patella deviates laterally when the knee reaches terminal extension; the Q-angle may be less accurate in full extension in a patient with a positive J sign.

At any point during the inspection portion of the examination, the clinician should assess for muscular atrophy, particular with respect to the vastus medialis obliquus (VMO), prior surgical incisions, and the presence/absence of an effusion.

Palpation

Many of the structures about the knee that may be pain generators can be directly palpated, including the patella itself, the infrapatellar fat pad, quadriceps and patellar tendon attachments on the patella, retinaculum/patellofemoral ligaments, tibial tubercle, iliotibial band, joint lines, and femoral condyles (among other structures). Palpating these structures can be helpful in determining an underlying anatomical source of pain, if one exists. In addition, palpating the patella during knee flexion/extension allows for an assessment of crepitus. In one study, 98% of adolescent patients with patellofemoral pain experienced pain with palpation over the MPFL.[23] Notably, larger body habitus as well as guarding can make the palpation portion of the examination more difficult.

Range of Motion and Strength

Knee ROM can be assessed in the seated, supine, or prone positions. Regardless of the position used, it is important to compare

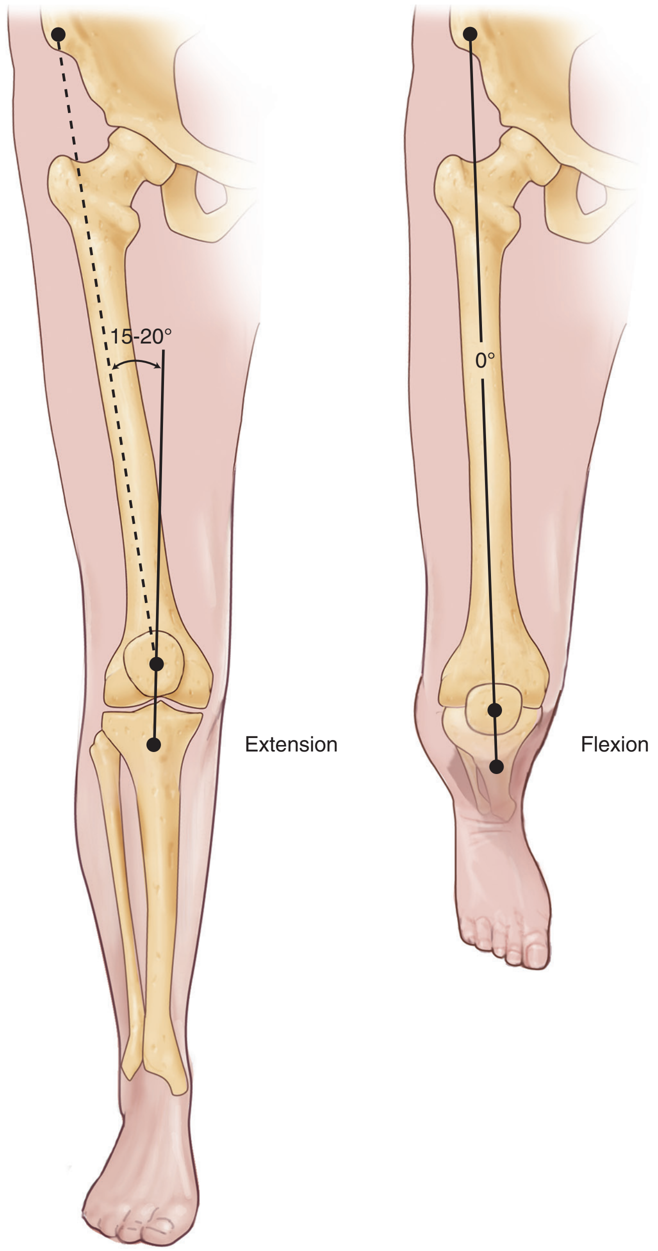

Fig. 106.5 Flexion of the knee decreases the Q angle as the result of internal tibial rotation. (From Tria Jr AJ, Klein KS. *An Illustrated Guide to the Knee.* New York: Churchill Livingstone; 1992.)

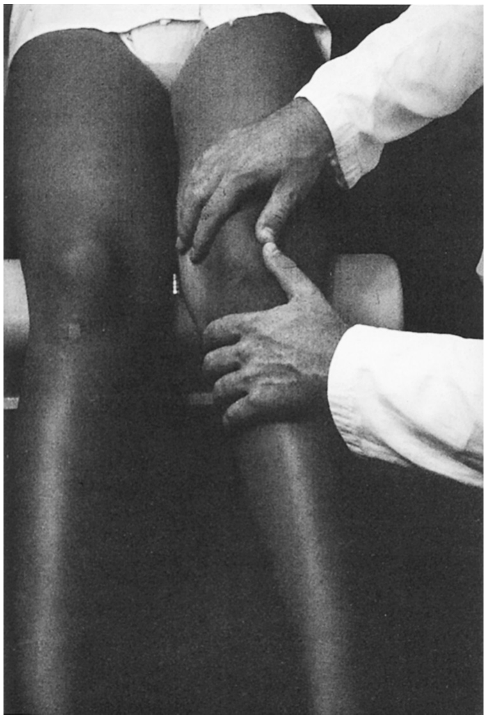

Fig. 106.6 The patellar compression test. With the knee flexed slightly to engage the patella in the femoral trochlea, direct compression is applied to the patella. (From Scott WN, ed. *Insall & Scott Surgery of the Knee.* 5th ed. Philadelphia: Elsevier; 2012.)

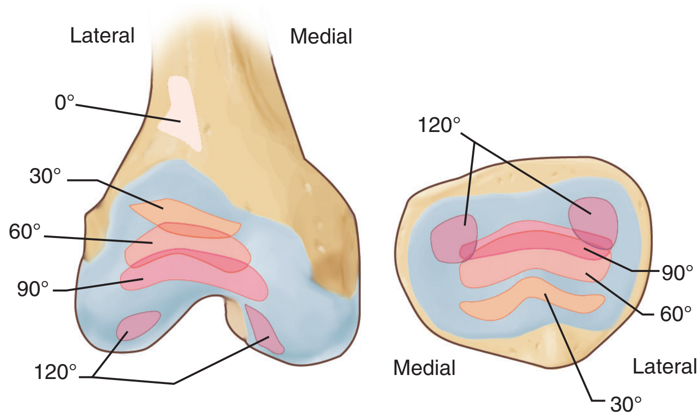

Fig. 106.7 The patellofemoral joint contact areas according to the knee flexion angle. (From Scott WN, ed. *Insall & Scott Surgery of the Knee.* 5th ed: Philadelphia: Elsevier; 2012.)

the symptomatic knee to the uninvolved knee, and to assess for baseline hyperextension. During ROM assessment, strength to resisted knee flexion and extension can also be assessed and compared side to side.

Provocative Tests

Provocative tests about the patella generally fall into two categories: tests to elicit pain, and tests to provoke instability. A variety of different tests have been described in which pressure is applied to the patella, including the patellar grind test, patellar compression test, and Clarke sign.[25,26] These tests involve some variation of applying pressure on the superior aspect of the patella with the examiner's thumb, often while asking the patient to contract their quadriceps muscle. If pain occurs, these tests are considered positive for retropatellar chondral pathology. A more gently applied posterior force to the patella during flexion and extension may elicit crepitus and pain in patients with retropatellar chondral pathology (Fig. 106.6). This test may help identify the location of pathology, because a more distal lesion will be painful in early flexion as the patella engages the trochlea, whereas a more proximal lesion will be painful at greater degrees of knee flexion (Fig. 106.7).

Patellar instability testing is discussed in detail in Chapter 105. The patellar apprehension test is performed by application of a laterally directed pressure to the medial aspect of

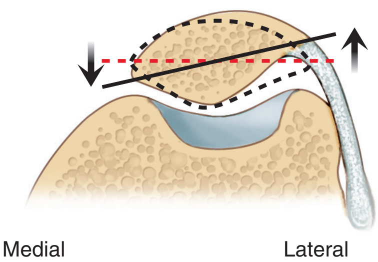

Fig. 106.8 The passive patellar tilt test. In full extension, the transverse axis of a normal patella tilts beyond the horizontal. The inability to perform this maneuver may indicate an excessively tight lateral retinaculum. (From Scott WN, ed. *Insall & Scott Surgery of the Knee*. 5th ed. Philadelphia: Elsevier; 2012.)

the patella with the knee in approximately 20 to 30 degrees of flexion. The test is positive if the patient experiences apprehension that the patella may dislocate laterally. If the patient experiences no apprehension but his/her patella is able to be subluxated or substantially translated laterally, the patient may have baseline ligamentous laxity and the clinician should examine the opposite knee for similar findings. Notably, while patients with patellofemoral pain may experience pain during apprehension testing, pain itself does not represent a positive test for instability. Manipulation of the patella in the transverse plane may reveal a tight retinaculum, as evidenced by a lack of motion (Fig. 106.8). The patellar tilt test should also be performed as an assessment for a tight retinaculum and may have some utility in determining possible interventions.[28] The Ober test is useful to assess for iliotibial (IT) band tightness that can contribute to anterolateral knee pain as well as patellar maltracking.[128]

Diagnostic Studies

Imaging studies, including radiographs, computed tomography (CT), and magnetic resonance imaging (MRI) can be helpful in the valuation of patients with patellofemoral pain. Radiographs, including anterior-posterior (AP), lateral, and axial (merchant or sunrise), are useful for evaluating the osseous morphology of the patellofemoral and tibiofemoral joints, as well as for identifying signs of arthritis. The lateral radiograph is perhaps the most useful in the evaluation of the patellofemoral joint, as it is useful for determining patellar height (Fig. 106.9) as well as trochlear dysplasia. Patella height can be determined via several different techniques, including the Caton-Deschamps index, Insall-Salvati ratio, Blackburne-Peel ratio, and the relationship of Blumensaat line to the inferior pole of the patella at 30 degrees of knee flexion. Patients with patella alta may have a predisposition to instability, whereas patients with patella baja may encounter increased patellar load and the subsequent risk of degenerative conditions. The lateral view is also helpful for identifying abnormal trochlear morphology according to the Dejour classification system.[28a,28b] In the Dejour system, trochlea dysplasia is classified as types A, B, C, or D (most severe), depending on if a crossing sign, supratrochlear spur, and/or double contour are present. Axial views such as the Merchant view (Fig. 106.10)[29] can help determine sulcus angle as well as the depth of the groove. Murray

Fig. 106.9 Lateral radiograph of the knee.

et al.[30] demonstrated that the lateral and axial views are the most useful views for evaluating instability and noted a high sensitivity of the lateral view for detecting prior dislocation; thus a normal lateral radiograph can help rule out instability as a component of anterior knee pain in some patients. The axial view is also useful for evaluating patellar tilt and subluxation. Different parameters have been defined, and the clinical utility of these parameters remains uncertain (Fig. 106.11). When evaluating axial radiographs, it is important to note that the sulcus of the trochlea lies lateral to the midline of the femoral condyles and is not completely centered as previously thought.[31] Interestingly, articular congruity may exist on magnetic resonance arthrogram (MRA) imaging, even when osseous incongruity appears on plain radiographs.[32]

Advanced imaging with CT can be very helpful given its ability to analyze the three-dimensional relationships of the patellofemoral joint. Data from CT imaging can help identify patellar tilt and tibial tuberosity–trochlear groove (TT-TG) value. Notably, indications for CT imaging in patients without patellar instability are limited, as patellar tilt and TT-TG measurements are most useful in the evaluation of patients experiencing patellar subluxation or dislocation events (see Chapter 105 for details on these measurements).

MRI is the imaging modality of choice for evaluating the articular cartilage and underlying subchondral bone and soft tissue structures such as the MPFL (Fig. 106.12). One study demonstrated that edema in the superolateral fat pad on MRI correlates with other anatomic parameters (e.g., trochlear morphology and patellar alignment) suggestive of patellar maltracking or impingement in the young, symptomatic patient.[33] Data points helpful for evaluating patellar instability, including TT-TG (originally described on CT scan) and TT-posterior cruciate ligament (TT-PCL), can be taken on MRI. In addition, lateral trochlear

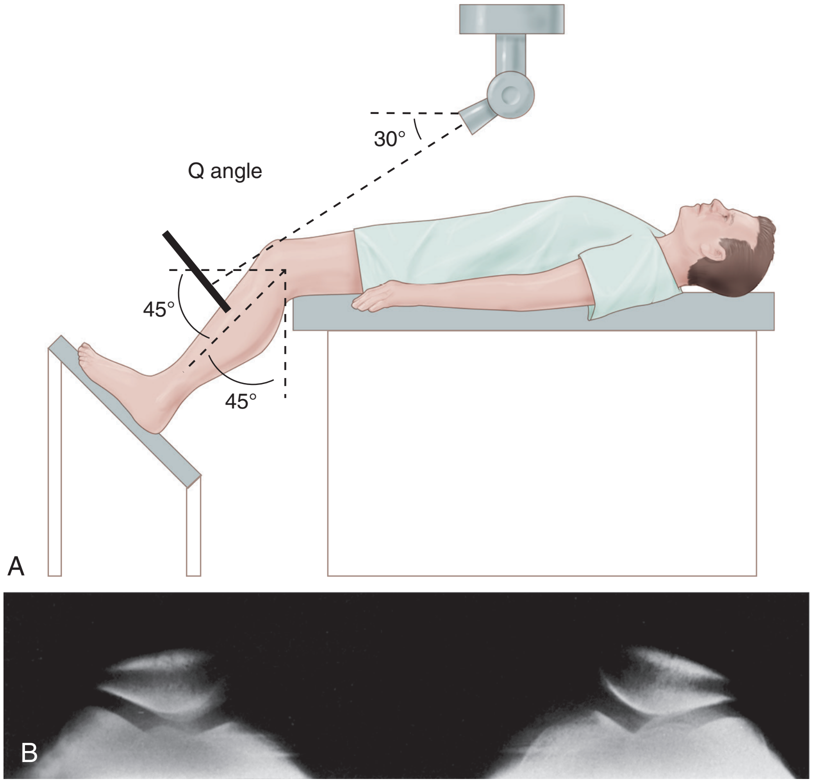

Fig. 106.10 Merchant view. (A) Technique. (B) A normal Merchant view. Patellofemoral alignment is normal bilaterally, and the osseous structures and articular cortices are normal. (From Scott WN, ed. *Insall & Scott Surgery of the Knee.* 5th ed. Philadelphia: Elsevier; 2012.)

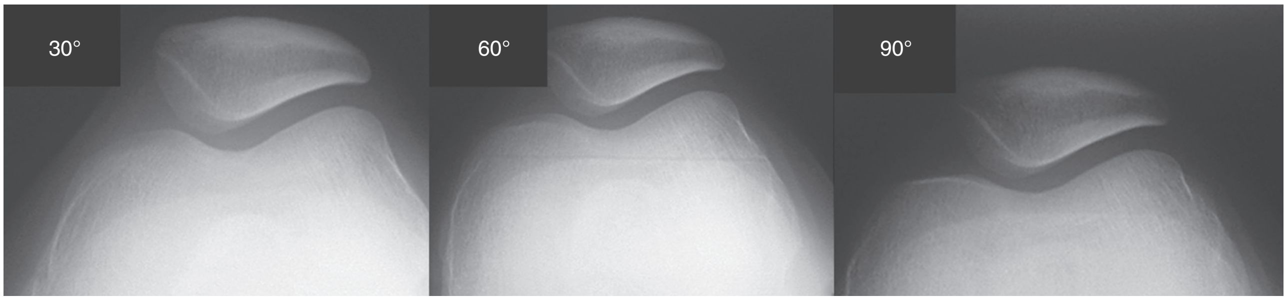

Fig. 106.11 Axial views performed at 30, 60, and 90 degrees of knee flexion. Note how the trochlear shape changes as flexion increases and the medial facet seems bigger. It is not necessary to routinely perform 60- and 90-degree flexion radiographs in common patellofemoral disorders. (From Scott WN, ed. *Insall & Scott Surgery of the Knee,* 5th ed. Philadelphia: Elsevier; 2012.)

inclination and medial trochlear inclination measurements can be taken on MRI, which may be more accurate than the sulcus angle in identifying abnormal trochlea morphology.

SPECIFIC CONDITIONS

Idiopathic Anterior Knee Pain/Patellofemoral Pain Syndrome

As mentioned throughout this chapter, as well as many other chapters in this text, a variety of factors contribute to anterior knee pain. As such, the differential diagnosis for anterior knee pain is broad (see Table 106.1). Many of the specific pathologies described in Table 106.1 have been discussed in detail throughout this text. Many patients have no identifiable lesion responsible for their patellofemoral pain, and the remaining portion of this chapter will focus on those pathologies. This type of condition is often referred to as PFPS. In patients with PFPS, the practitioner may find some potentially predisposing factors on physical examination or imaging, including malalignment (lower limb or patella), muscular imbalance and/or atrophy, and overuse. In a prospective study, patients with a hypermobile patella had a higher incidence of anterior knee pain, and in a separate study this examination finding correlated with a worse prognosis for patients with patellofemoral pain.[34,35] However, these findings can be common in patients without anterior knee pain. Overall, this is a diagnosis of exclusion, and the lack of objective, diagnostic criteria make it challenging for determining optimal treatment options.

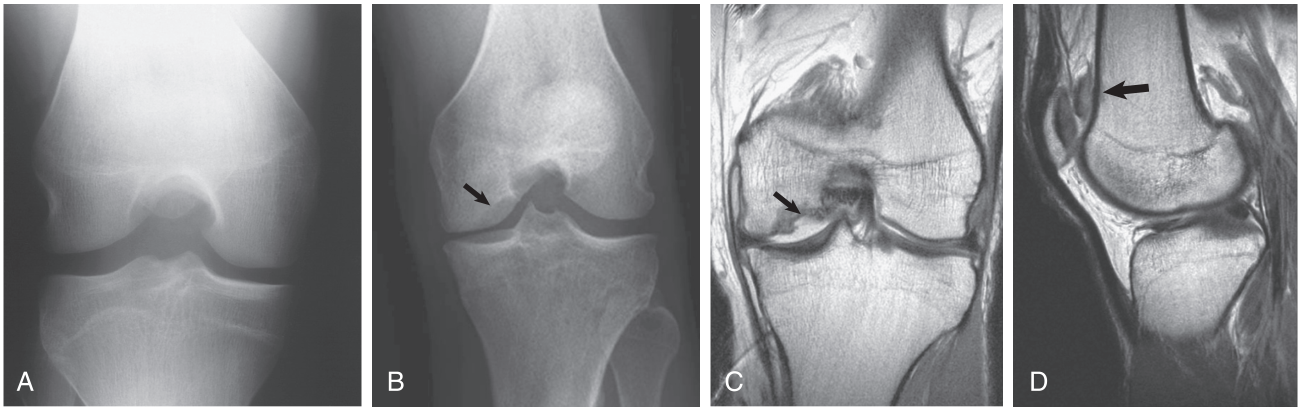

Fig. 106.12 Tunnel view. (A) A normal tunnel view demonstrates the posterior aspect of the femoral condyles, the tibial spines, the articular surfaces of the tibial plateau, and the intercondylar notch. (B) A tunnel view from a different patient demonstrating an ovoid area of lucency at the inner margin of the medial femoral condyle *(arrow)* that is suspicious for osteochondritis dissecans. Coronal (C) and sagittal proton density (D) magnetic resonance imaging confirms the large osteochondral defect *(arrow)* with a completely displaced osteochondral fragment located in the suprapatellar joint recess *(arrow)*. (Case courtesy the Hospital for Special Surgery, New York. From Scott WN, ed. *Insall & Scott Surgery of the Knee.* 5th ed. Philadelphia: Elsevier; 2012.)

The treatment of idiopathic PFPS focuses on primarily activity modification, physical therapy and muscular strengthening, bracing/taping, orthotics, electrotherapy, and pharmacotherapy. A systematic review of exercise therapy for patellofemoral pain demonstrated that exercise therapy showed little benefit with respect to pain reduction when compared to no exercise therapy.[37] In addition, another systematic review of the use of therapeutic ultrasound for treating PFPS failed demonstrate any clinically important effect on pain relief.[129] Furthermore, Callaghan et al. reviewed the use of patellar taping for PFPS in adults and found that there was limited evidence to suggest that taping can significantly improve outcomes compared to other exercise-based interventions not incorporating taping.[130] One study looking at military recruits with no history of anterior knee pain who were prospectively divided into a bracing group (dynamic patellofemoral brace) and a non-bracing group during basic training showed an almost 20% decrease in anterior knee pain symptoms in the bracing group (18.5% experienced pain versus 37% in the non-bracing group).[131] Another study showed that patients with a pronated foot type and poorer footwear motion control properties who wore a foot orthoses were associated with reduced anterior knee pain during the single-leg squat and improvements in the number of pain-free single-leg rises from sitting.[132] A systematic review of pharmacotherapy for the treatment of anterior knee pain found that there is only limited evidence for the effectiveness of nonsteroidal antiinflammatory drugs (NSAIDs) for short-term pain reduction.[38] Bizzini et al. analyzed the quality of randomized controlled trials for treatment of patellofemoral pain and showed that acupuncture, quadriceps strengthening, the use of a resistive brace, and the combination of exercises with patellar taping and biofeedback were effective in decreasing pain and improving function.[133] Overall, there is mixed evidence regarding the success of nonoperative treatment for the management of patellofemoral pain.

For patients who have significant disability despite use of nonoperative treatment modalities for a minimum of 6 months, arthroscopy can be considered for diagnostic and therapeutic reasons, including assessment of potential chondral lesions (please see Chapters 96 and 104 on patellar cartilage disorders). It can be helpful before performing arthroscopy to attempt a diagnostic intra-articular injection with local anesthesia (with or without corticosteroid); if such an injection does not provide even temporary relief, the benefits of an arthroscopy are less predictable. Patellar denervation has been described using electrocautery[39] and can be accomplished via an anterolateral, anteromedial, and superior portal (either medial and lateral). This procedure should be applied sparingly, and the authors strongly urge the clinician to search for other diagnoses before labeling a condition as idiopathic anterior knee pain or PFPS.

Synovial Impingement Syndromes

Impingement is a common cause of patellofemoral pain. The offending structures are typically a plica or the infrapatellar fat pad. The plicae are embryologic remnants that are normal anatomic structures. Differentiating between pathologic and normal plica is difficult.[40] The medial plica is most commonly symptomatic (Fig. 106.13).[41] This plica was shown to be present in approximately 80% of the population in one study series.[42] Patients with synovial plica syndrome (SPS) often describe activity-related pain that worsens with flexion and is relieved with extension. With a hypertrophic plica, a thick palpable cord may be palpable on physical examination. Both flexion and extension tests have been described to aid in the diagnosis of pathologic plicae, but they are of limited utility.[43] MRI may detect a prominent plica, but no study has demonstrated a role for determining pathologic versus normal plicae. Chondral pathology of the medial femoral condyle has been noted in cases of severe plicae, and this pathology may be appreciated on MRI.[44] The treatment of symptomatic plicae should begin with nonoperative options, including activity modification, NSAIDs, and if needed, intra-articular corticosteroid injections. For patients who do not respond to nonoperative treatment, arthroscopy with plica excision is a

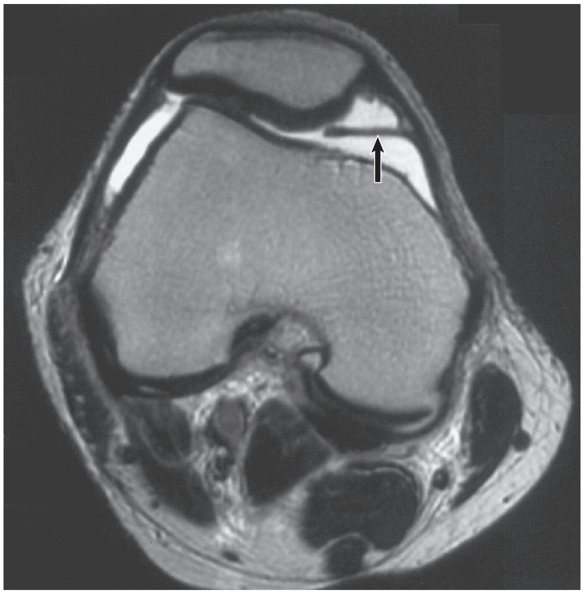

Fig. 106.13 An axial magnetic resonance image with a low-signal, thick medial patellar plica *(arrow)* that is highlighted by the large effusion. (From Scott WN, ed. *Insall & Scott Surgery of the Knee.* 5th ed. Philadelphia: Elsevier; 2012.)

relatively straightforward option with low associated morbidity, and in patients with associated cartilage degeneration, midterm results support plica excision.[46] If proceeding to arthroscopy, the surgeon should include a thorough evaluation for associated concomitant knee pathology.

The infrapatellar fat pad was implicated in anterior knee pain by Hoffa[47] in 1904. As previously described, this area has a rich innervation and can result in substantial anterior knee pain, associated with superficial swelling and tenderness to palpation. The Hoffa maneuver involves applying pressure just medial and lateral to the patellar tendon (separately) and extending the knee, which may elicit either pain or apprehension. While the diagnosis is primarily clinical, some evidence indicates that MRI may aid in the diagnosis, and use of MRI may become more relevant with the utilization of MRIs with stronger magnets.[33,48] The treatment of fat pad impingement is similar to the initial treatment for symptomatic plica, including nonoperative strategies. For patients who do not respond to conservative modalities, surgical excision of the fat pad has been described.[48–50] The authors recommend this treatment only when the patient has exhausted all nonoperative modalities and have excluded any other factor that may contribute to the patient's symptoms (alignment, muscle imbalance, etc.).

Intraosseous Hypertension of the Patella

Increased intraosseous pressure has been described in the femur and tibia of painful knees by Arnoldi and colleagues.[51] These investigators later correlated knee pain with increased pressure in the patella. It has been suggested that this phenomenon may be related to decreased venous outflow based on intraosseous phlebography.[52] The diagnostic criterion for patellar hypertension is an increase in intraosseous pressure of greater than 25 mm Hg with sustained knee flexion. Schneider and colleagues[53,54] reported on a series of patients with patellofemoral pain that failed to respond to conservative measures. After administration of a local anesthetic a "provocation test" was performed (i.e., the reproduction of symptoms by raising the intraosseous pressure), and patients underwent intraosseous drilling and decompression. At 1-year follow-up, 88% of subjects had an objective decrease in pressure measurement.[53] A subsequent publication demonstrated improved clinical outcome scores (the visual analog scale score decreased from 7.6 to 2.1) with this treatment protocol at 3 years postsurgery.[55] While overall this diagnosis is somewhat controversial, several series have demonstrated that patellar drilling and decompression may benefit patients who fail to respond to nonoperative treatment for patellofemoral pain. It remains unclear if this diagnosis is the primary pathology or if it is secondary to other disease processes (including lateral patella compression syndrome).

Lateral Patella Compression Syndrome

Ficat[56] originally described lateral patellar compression syndrome, and the underlying etiology is thought to be attributed to a tight lateral retinaculum. Patients typically report pain (without instability) while ascending or descending stairs and may have a positive theatre sign (i.e., pain with prolonged sitting). As the condition progresses, chondral wear can occur, and symptoms including effusions, locking, clicking, catching, and/or giving way may occur. On examination, patients will have pain with patellar compression testing, particularly along the lateral aspect. Notably, patients will not exhibit excessive patellar mobility as the retinaculum is tight, and the examiner may have difficulty everting the lateral patellar edge above parallel to the floor. These patients may also have subtle findings of an increased Q-angle, a tight iliotibial band (IT band, seen with a positive Ober test[128]), and increased foot pronation. Axial radiographs or axial CT scans will demonstrate increased lateral patellar tilt and a decreased lateral patellofemoral angle. The treatment of lateral patellar compression syndrome should primarily focus on nonoperative modalities to decrease inflammation, including the use of NSAIDs, activity modification, and physical therapy. Modalities to help loosen the lateral retinaculum and IT band are useful adjuncts. For patients who fail to respond to nonoperative treatment and continue to have pain and dysfunction, surgery can be considered. This condition remains one of the few indications for an isolated lateral release. Certainly, if performing a lateral release, it is important to ensure the release is not too aggressive to avoid the development of iatrogenic medial patellar instability.[57] If any evidence of patellar hypermobility exists, then an open lateral retinacular lengthening procedure can be considered.

Complex Regional Pain Syndrome

Reflex sympathetic dystrophy was suggested as a cause of some anterior knee pain by Merchant[10] in 1988. Specific to anterior knee pain, scintigraphy has been demonstrated to aid in the diagnosis of sympathetically mediated pain, and a sympathetic blockade may be a useful treatment modality.[120] The discussion of complex regional pain syndrome is beyond the scope of this chapter, but it is an important consideration in the differential diagnosis for chronic patellofemoral pain.

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For a complete list of references, go to ExpertConsult.com.

SELECTED READINGS

Citation:
Dye SF. The pathophysiology of patellofemoral pain: a tissue homeostasis perspective. *Clin Orthop Relat Res*. 2005;436:100–110.

Level of Evidence:
V

Summary:
In this article, the authors evaluate the perspective that tissue homeostasis such as intraosseous hypertension is the etiology of patellofemoral pain in contrast to structural abnormalities.

Citation:
Dye SF, Vaupel GL, Dye CC. Conscious neurosensory mapping of the internal structures of the human knee without intraarticular anesthesia. *Am J Sports Med*. 1998;26(6):773–777.

Level of Evidence:
V

Summary:
In this article, the authors report the case of a physician who underwent knee arthroscopy without use of an anesthetic, and the degree of noxious stimulus in various locations is noted.

Citation:
Kodali P, Islam A, Andrish J. Anterior knee pain in the young athlete: diagnosis and treatment. *Sports Med Arthrosc*. 2011;19(1):27–33.

Level of Evidence:
V

Summary:
In this review article, the various etiologies and treatments of anterior knee pain are discussed.

Citation:
Pihlajamaki HK, Visuri TI. Long-term outcome after surgical treatment of unresolved Osgood-Schlatter disease in young men: surgical technique. *J Bone Joint Surg Am*. 2010;92A(suppl 1 Pt 2):258–264.

Level of Evidence:
IV

Summary:
In this article, a case series is described with follow-up of patients with Osgood-Schlatter disease who did not respond to conservative management and subsequently underwent operative intervention. In this series, 87% of subjects had no activity limitation with activities of daily living and 75% returned to their preoperative athletic level.

Citation:
Witvrouw E, Lysens R, Bellemans J, et al. Intrinsic risk factors for the development of anterior knee pain in an athletic population. A two-year prospective study. *Am J Sports Med*. 2000;28(4):480–489.

Level of Evidence:
II

Summary:
In this description of a prospective cohort study, 282 athletic patients were followed up to evaluate risk factors for anterior knee pain. The authors noted a significant correlation between the incidence of patellofemoral pain and patients with a shortened quadriceps muscle, an altered vastus medialis obliquus muscle reflex response time, decreased explosive strength, and a hypermobile patella.

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